

tf.compat.v1.losses.huber_loss Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/losses/huber_loss

Adds a [Huber Loss](https://en.wikipedia.org/wiki/Huber_loss) term to the training procedure.

Function Signatures

```
tf.compat.v1.losses.huber_loss( labels, predictions,  
weights=1.0, delta=1.0, scope=None,  
loss_collection=ops.GraphKeys.LOSSES,  
reduction=Reduction.SUM_BY_NONZERO_WEIGHTS )  

$$0.5 * x^2 \text{ if } |x| \leq d \quad 0.5 * d^2 + d * (|x| - d) \text{ if } |x| > d$$

```

Code Examples

```
tf.compat.v1.losses.huber_loss(  
    labels,  
    predictions,  
    weights=1.0,  
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    scope=None,  
    loss_collection=ops.GraphKeys.LOSSES,  
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)
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0.5 * x^2                if |x| <= d  
0.5 * d^2 + d * (|x| - d) if |x| > d
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0.5 * x^2                if |x| <= d  
0.5 * d^2 + d * (|x| - d) if |x| > d
```

Documentation

Adds a Huber Loss term to the training procedure.

For each value x in $\text{error} = \text{labels} - \text{predictions}$, the following is calculated:

where d is delta.

`weights` acts as a coefficient for the loss. If a scalar is provided, then the loss is simply scaled by the given value. If `weights` is a tensor of size `[batch_size]`, then the total loss for each sample of the batch is rescaled by the corresponding element in the weights vector. If the shape of `weights` matches the shape of predictions, then the loss of each measurable element of predictions is scaled by the corresponding value of `weights`.

Returns

Raises

eager compatibility

The `loss_collection` argument is ignored when executing eagerly. Consider holding on to the return value or collecting losses via a `tf.keras.Model`.